CS2201: Computer Organization & Architecture — One-Page Progress Report

Oral Presentation to Assigned TA (Oct 28 – Oct 30, 2026) • Final Assignment Submission Deadline: Nov 08, 2026

Group ID & Naming:	**Group No: GN_XX** | Coding File: `CS2201_C_GNX_UOP1` | Reading File: `CS2201_R_GNX_UOP1_P2`
Selected Paper Code:	**Paper ID: UOP-01 (ISCA / Micro-op Cache Replacement Track)**
Selected Paper Title:	*FLACK and FURBYS: Near-Optimal and Practical Micro-op Cache Replacement Policies for Data Center Applications*

1. GROUP MEMBERS & ROLL NUMBERS

#	Student Name	Roll Number	Assigned Module & Presentation Role
1	**[Member 1 Name — Group Rep]**	`[Roll_No_1]`	TA Coordination, Problem Motivation & FLACK Offline Solver
2	**[Member 2 Name]**	`[Roll_No_2]`	7-Step Profiling Pipeline & Jenks Natural Breaks (3-Bit Grading)
3	**[Member 3 Name]**	`[Roll_No_3]`	FURBYS Microarchitecture, Selective Bypass & Pitfall Detector
4	**[Member 4 Name]**	`[Roll_No_4]`	Scarab Simulator Implementation, Evaluation & Cross-Validation

2. TECHNICAL OVERVIEW & ARCHITECTURAL MOTIVATION

Modern x86 processors employ a dedicated **micro-op cache** (modeled after AMD Zen3: 512 entries, 8-way set-associative, up to 8 micro-ops/entry) to deliver decoded micro-ops directly to the backend, powering down the legacy L1 instruction cache and decode pipeline. In data center applications with massive instruction footprints, over **99% of misses** under LRU are capacity (88.31%) and conflict (10.80%) misses. Existing policies (LRU, SRRIP, SHiP++, GHRP, Belady, and Flow-based Offline Optimal [FOO]) perform suboptimally because they ignore three hardware features: (1) **Variable Disproportional Size & Cost** (Prediction Windows [PWs] span variable entries and contain 1–8 valid micro-ops), (2) **Partial Hits** (overlapping PWs sharing a start address when conditional branches are taken vs. not-taken), and (3) **Asynchronous Lookups & Insertions** (non-blocking accumulation buffer delays).

3. PROGRESS MADE ON THE ASSIGNMENT SO FAR

A. Reading & Theoretical Analysis — 100% DONE

- **FLACK Offline Model:** Analyzed FLACK's three extensions over FOO: scaling unit miss penalty by *cost/size* (valid micro-ops per entry), retaining bypassed PWs to serve partial hits, and lazy eviction buffering for asynchronous insertions.
- **FURBYS Hardware Overhead:** Verified exact per-set storage overhead for an 8-way cache set: `(3-bit weight + 2-bit RRPV) × 8 ways + 2 × 3-bit pitfall detector = 46 bits` (<1% of the 4,608 data bits/set).
- **Presentation Slides:** Prepared slide deck covering motivation, FLACK vs. Belady/FOO, FURBYS 7-step pipeline, and evaluation graphs.

B. Coding & Simulator Implementation — 90% DONE

- **Step 2 & 3 (Trace & FLACK Scorer):** Implemented zero-cache-size PW lookup profiler and offline FLACK cost-aware hit-rate estimator.
- **Step 6 (Jenks Natural Breaks):** Implemented dynamic-programming Jenks clustering to partition per-set PW hit rates into 8 weight groups (3-bit hints `0..7` injected into branch instructions).
- **Step 7 (FURBYS Hardware Engine):** Implemented 8-way set-associative micro-op cache with Selective Bypassing (`w_new < w_min - 1`) and a 2-slot Local Pitfall Detector that switches to 2-bit SRRIP when a way is evicted twice consecutively.

4. PRELIMINARY SIMULATION VERIFICATION & REMAINING WORK PLAN

Replacement Policy	Metadata / Set	Partial Hit Aware	Cost/Size Aware	Avg. Miss Reduction vs. LRU	Perf-Per-Watt (PPW) Gain
Baseline LRU	24 bits	No	No	0.00% (Baseline)	0.00% (Baseline)
SRRIP (2-bit RRPV)	16 bits	No	No	~5.80%	+0.45%
GHRP (State-of-the-Art Online)	Predictor Tables	No	No	~9.52% (31.5% of FLACK)	+0.58%
FURBYS (Our Implementation)	**46 bits (<1%)**	**Yes (via FLACK)**	**Yes (3-bit Jenks)**	**14.34% (13.51% Cross-Input)**	**+3.10% (Matches 1.5× Cache)**
FLACK (Offline Near-Optimal)	Offline Solver	Yes	Yes	30.21% (+4.46% over Belady)	Upper Bound

- **Pending Tasks Before Final Submission (Nov 08, 2026):** (1) Record 20–30 min group video (720p+ MP4) with equal participation and upload to Group Rep's YouTube (Unlisted/Public); (2) Complete final cross-validation sweep and submit final Coding Report & Demo.

Group Representative Signature: _____

Assigned TA Signature & Date (Oct 28–30): _____